

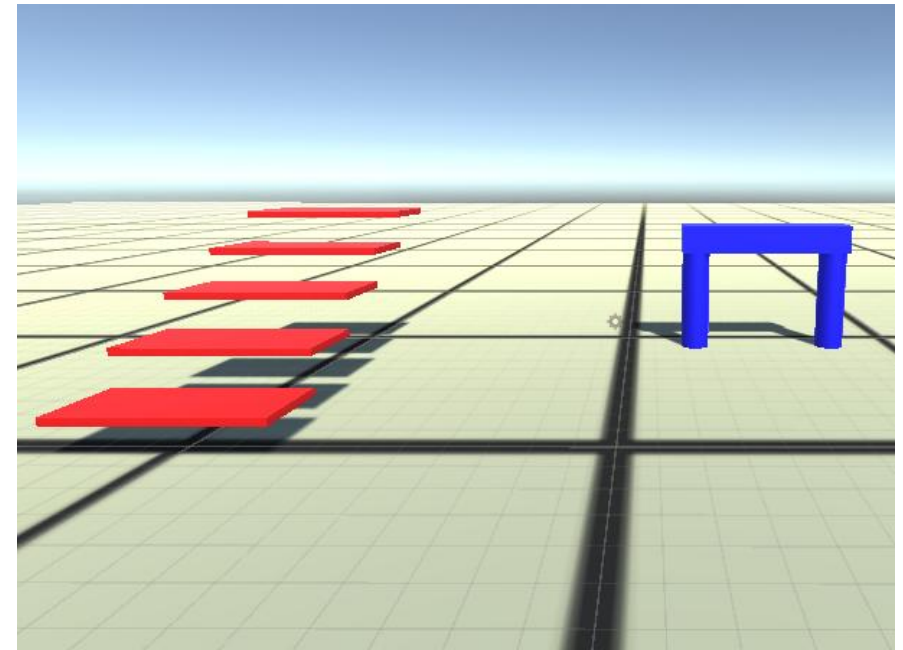
Video Game Programming: Exercises

E01

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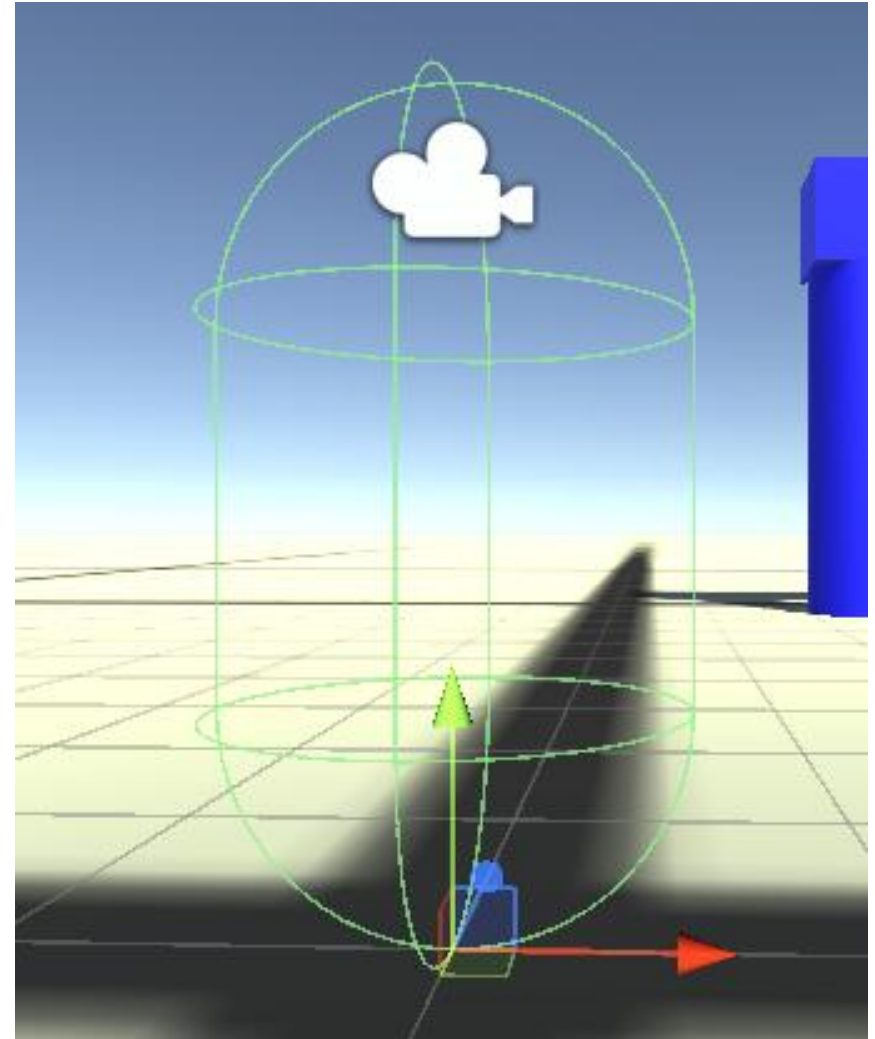
Transforms and materials

- In the scene E01-1, use “3D Object” menu and edit position, rotation, and scale of Transform components to make the platforms and arc in fig.
- Create a script called “PlatformMovement” to move the platforms up/down, left/right or back/forward according to direction, maxDistance, and speed parameters
- Assign materials to the scene so that it looks like in fig.



Character controller

- In the scene E01-2, create a first-person character object with a character controller
- Add a camera as a child as in fig.
- Implement character rotation on Y axis and camera rotation on X axis with Look action
- Implement character movement in the current forward direction with Move action
- Add the script to apply forces on successful raycast



Physics and prefabs

- In the scene E01-3, spawn robots that are moved using physics
- Implement jumping (and landing on ground) for the first-person character controller with the Jump action
- Add kinematic rigid bodies to platforms
- Add a trigger collider to the arc and destroy objects passing through it
- Make prefabs for the platform set, arc, and first-person character

